MANAGEMENT INFORMATION SYSTEMS
ICT2162
LECTURE 01

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University of Ruhuna

Introduction

- Course Unit

  -Management Information Systems – ICT2162

- Credits

  -2

- Lecture Hours

  -30h

Course Unit Learning Outcomes

- After completing this course unit, students would be able to,

1. Identify organizations, Management, and the networked Enterprise.
2. Explore Information Technology infrastructure and security of an organization.
3. Explain key concepts related to operational excellence and customer intimacy.
4. Identify how to build and manage information systems.

Course Content

- Information Systems in Global Business Today
- Information Systems, Organizations, and Strategy
- Ethical and Social Issues in Information Systems
- IT Infrastructure and Emerging Technologies
- Foundations of Business Intelligence: Databases and Information Management
- Securing Information Systems
- Achieving Operational Excellence and Customer Intimacy: Enterprise Applications
- Managing Knowledge and Artificial Intelligence
- Enhancing Decision Making
- Building Information Systems
- Managing Global Systems

References

- “Management Information Systems: Managing the Digital Firm” by “K. C. Laudon, J. P. Laudon”
- “Essentials of MIS” by “K. C. Laudon, J. P. Laudon”

Evaluation Criteria

- **Eligibility**

80% attendance is **mandatory**

- **Continuous Assessment (CA) ….30%**

a. Assignments 20%
b. Quizzes 10%

- **Final Assessment (FA) on LO1 to LO4 ……70%**
- Theory 70%

Information Systems in Global Business Today

Objectives

- How are information systems transforming business, and why are they so essential for running and managing a business today?
- What is an information system?
- How does it work?
- What are its management, organization, and technology components?
- Why are complementary assets essential for ensuring that information systems provide genuine value for organizations?
- What academic disciplines are used to study information systems, and how does each contribute to an understanding of information systems?
- How will MIS help my career?

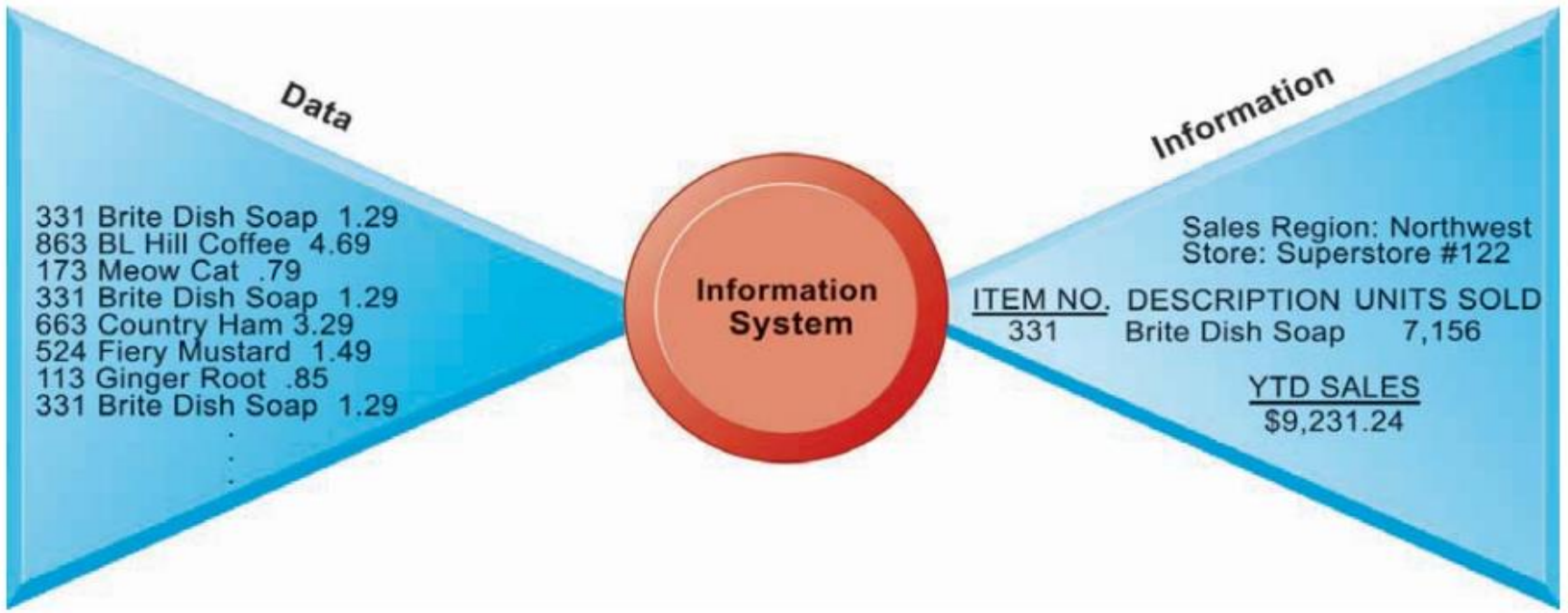
What is an Information System?

A set of interrelated components that

- collect, process, store and distribute information
- to support decision making and control in an organization.

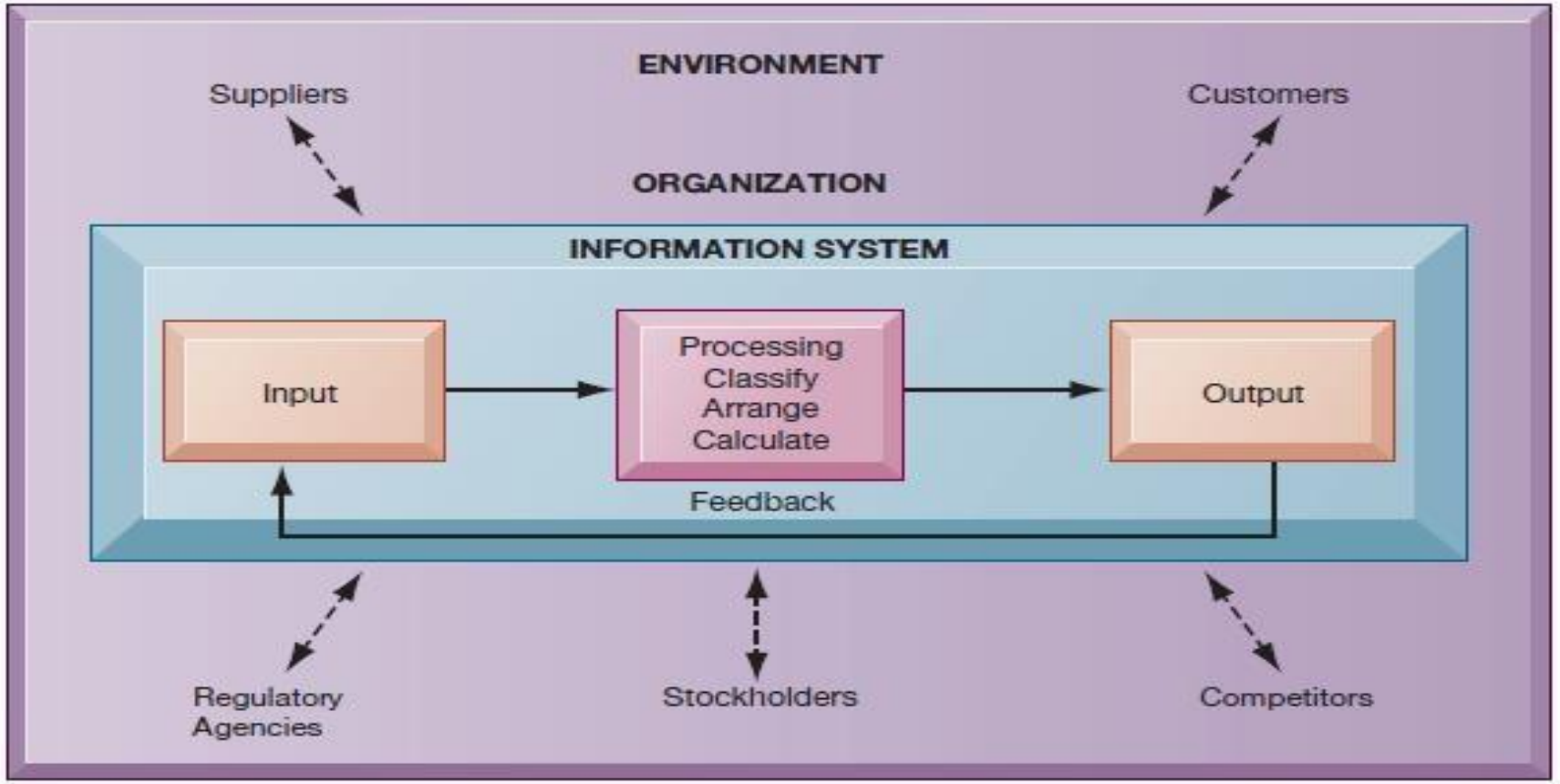
In addition to supporting decision making, coordination, and control, information systems may also help managers and workers analyze problems, visualize complex subjects, and create new products.

Data and Information



Raw data from a supermarket checkout counter can be processed and organized to produce meaningful information, such as the total unit sales of dish detergent or the total sales revenue from dish detergent for a specific store.

Functions of an Information System



An information system contains information about an organization and its surrounding environment. Feedback is output returned to appropriate people or activities in the organization to evaluate and refine the input. Environmental actors, such as customers, suppliers, competitors, stockholders, and regulatory agencies, interact with the organization and its information systems.

- Three activities in an information system

  - produce the information that organizations need to make decisions

  -control operations

  -analyze problems and create new products or services.
- These activities can be summarized as input, processing, and output.
- Input captures or collects raw data from within the organization or from its external environment.
- Processing converts this raw input into a meaningful form.
- Output transfers the processed information to the people who will use it or to the activities for which it will be used.

- Information systems also require feedback, which is output that is returned to appropriate members of the organization to help them evaluate or correct the input stage.

Example : Disney world’s system for controlling crowd.

Dimensions of Information Systems

- Organizations, Technology and Management



- **Organizations**

Information systems are an integral part of organizations.

For some companies, such as credit reporting firms, there would be no business without an information system.

The key elements of an organization are its people, structure, business processes, politics, and culture.

- **Management**

Organizations have a structure that is composed of different levels and specialties.

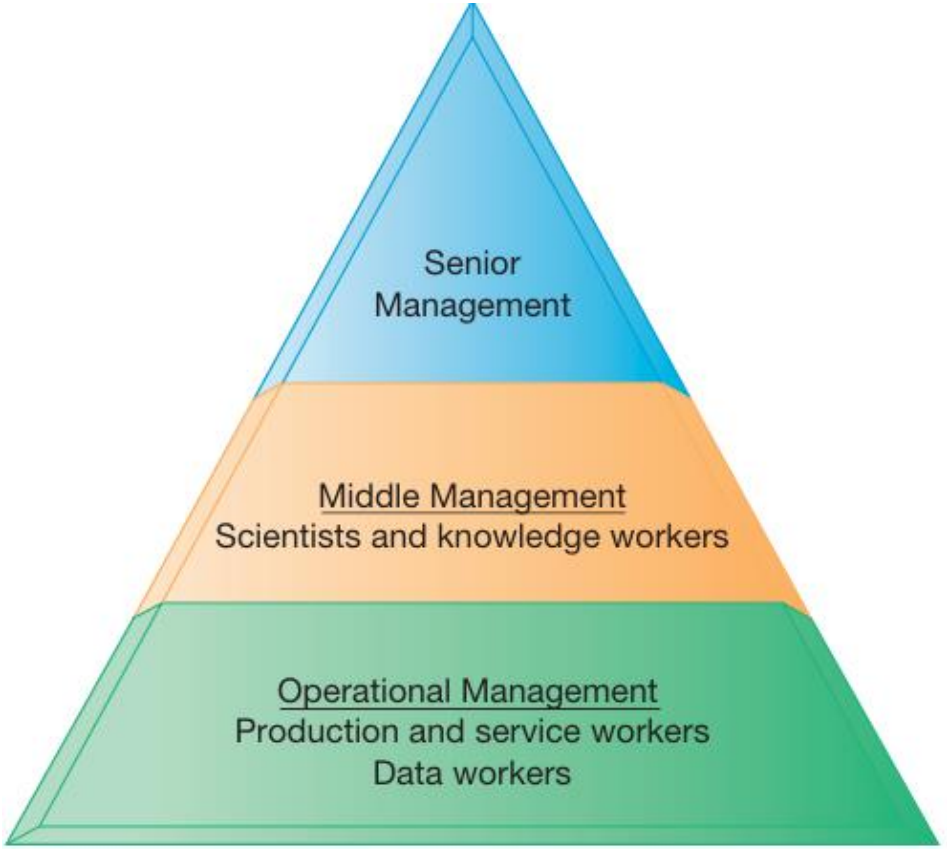
Their structures reveal a clear-cut division of labor. Authority and responsibility in a business firm are organized as a hierarchy, or a pyramid structure.

The upper levels of the hierarchy consist of managerial, professional, and technical employees, whereas the lower levels consist of operational personnel.

Senior management makes long-range strategic decisions about products and services as well as ensures financial performance of the firm.

Middle management carries out the programs and plans of senior management.

Operational management is responsible for monitoring the daily activities of the business.



- **Information Technology**

Computer hardware - physical equipment used for input, processing, and output activities in an information system.

Computer software - consists of the detailed, preprogrammed instructions that control and coordinate the computer hardware components in an information system.

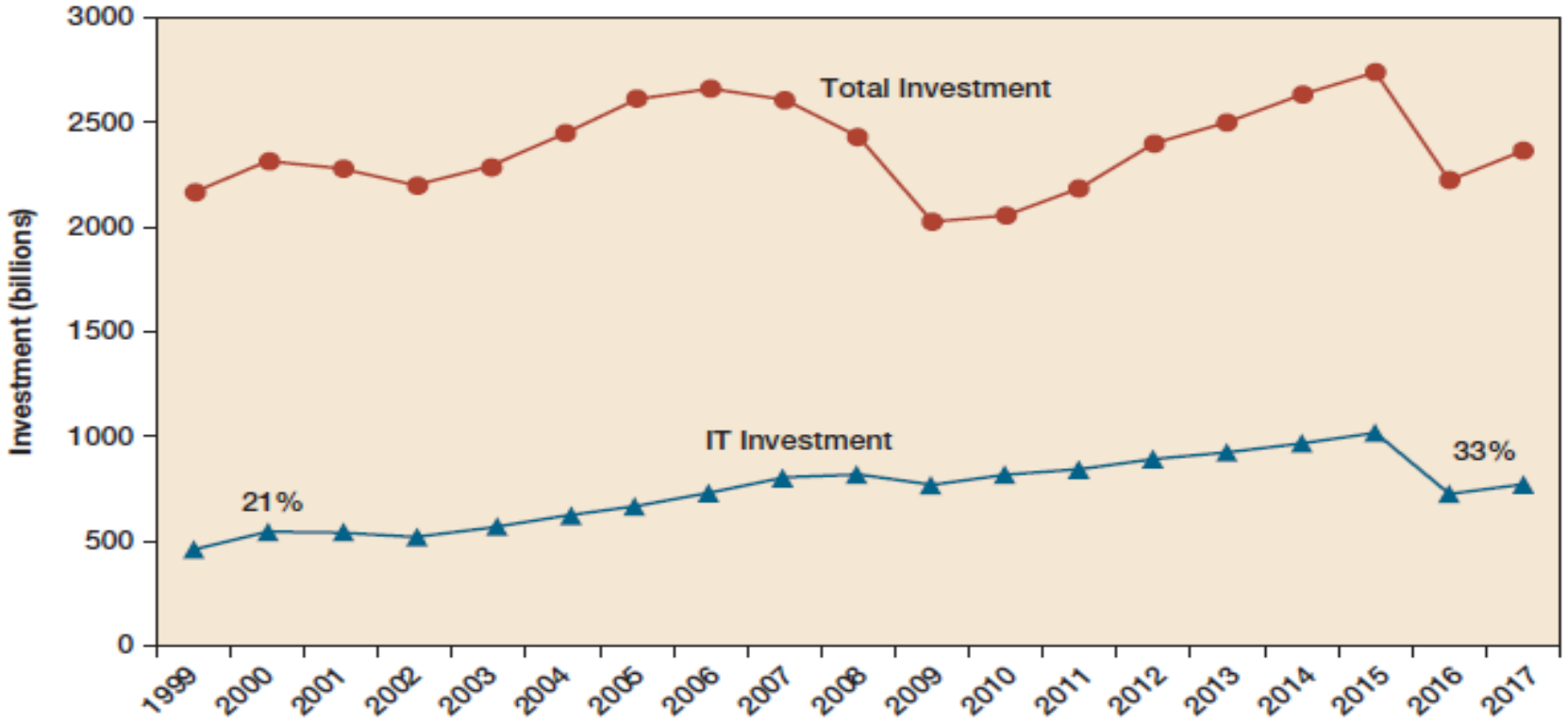
Data management technology - consists of the software governing the organization of data on physical storage media.

Networking and telecommunications technology - consisting of both physical devices and software, links the various pieces of hardware and transfers data from one physical location to another.

What's New in Management Information Systems

- IT Innovations
  Ex: Cloud computing (Google Drive, OneDrive)
- New Business Models
  Ex: Netflix
- E-commerce Expansion
- Management Changes
- Changes in Firms and Organizations

How Information Systems are Transforming Business



- Example: Mobile Phone Industry



How Information Systems are Transforming Business

- Increased technology investments
- IS provide economic value to business
- Increased responsiveness to customer demands
- Shifts in media and advertising and commerce
- New laws in IS

Globalization & Information Systems

- 80% of the toys sold in U.S are manufactured in China, while about 90% of the PCs manufactured in China use American made Intel or AMD chips.
- Not just goods move across boarders; so too do jobs
- 24 X 7 culture
- Internet reduces cost of operating on global scale
- Emergence of technology based companies.
  - Google
  - eBay
  - Amason.com


Meta
facebook

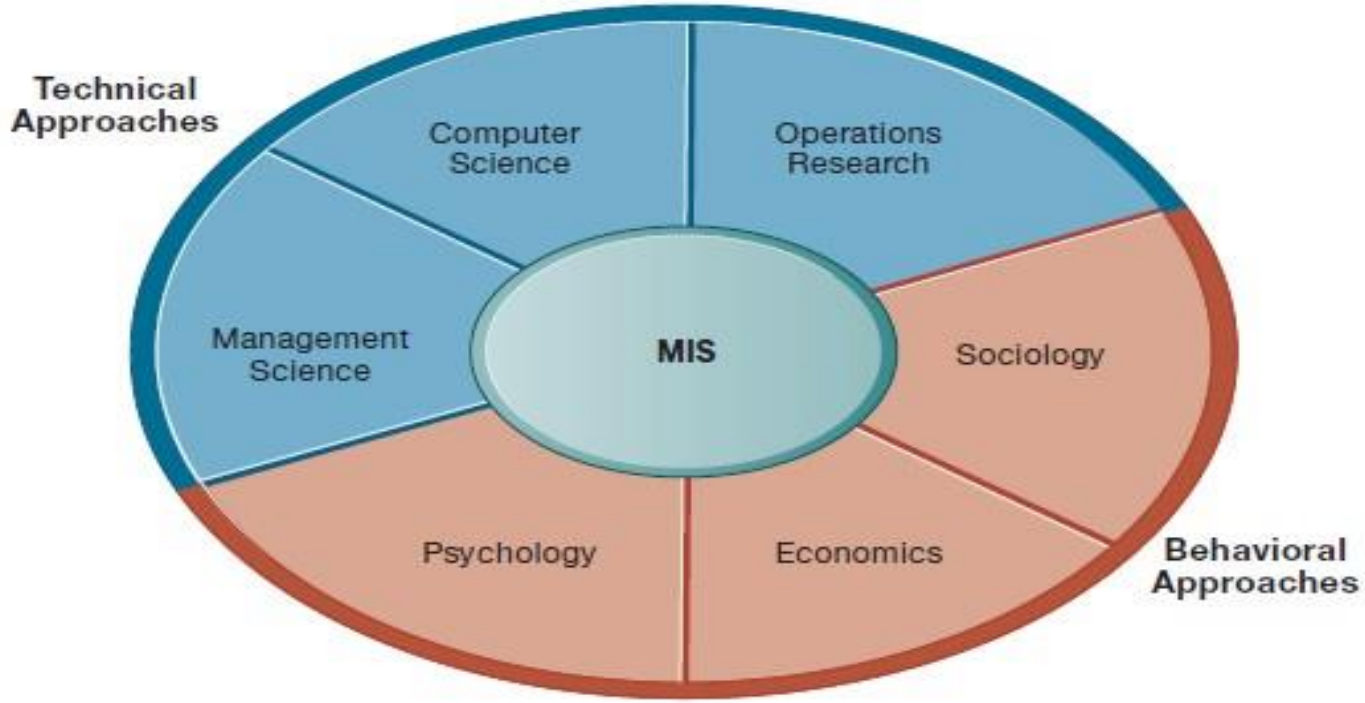
Emergence of Digital Firms

- Digital firm is one in which nearly all the organization's significant business relationships with customers, suppliers, and employees are digitally enabled and mediated.
- Any piece of information required to support key business decisions is available at anytime and anywhere in the firm.

Why Business Firms Invest Heavily on Information Systems?

- Business firms invest heavily in information systems to achieve six strategic business objectives.
  - Operational excellence
    - Wal-Mart
  - New products, services, and business models
    - Apple Inc.
  - Customer and supplier intimacy
    - The Mandarin Oriental hotel group
  - Improved decision making
  - Competitive advantage
    - Toyota
  - Survival
    - HSBC

Contemporary Approaches to Information Systems



- **TECHNICAL APPROACH**

The disciplines that contribute to the technical approach are computer science, management science, and operations research.

Computer science is concerned with establishing theories of computability, methods of computation, and methods of efficient data storage and access.

Management science emphasizes the development of models for decision-making and management practices.

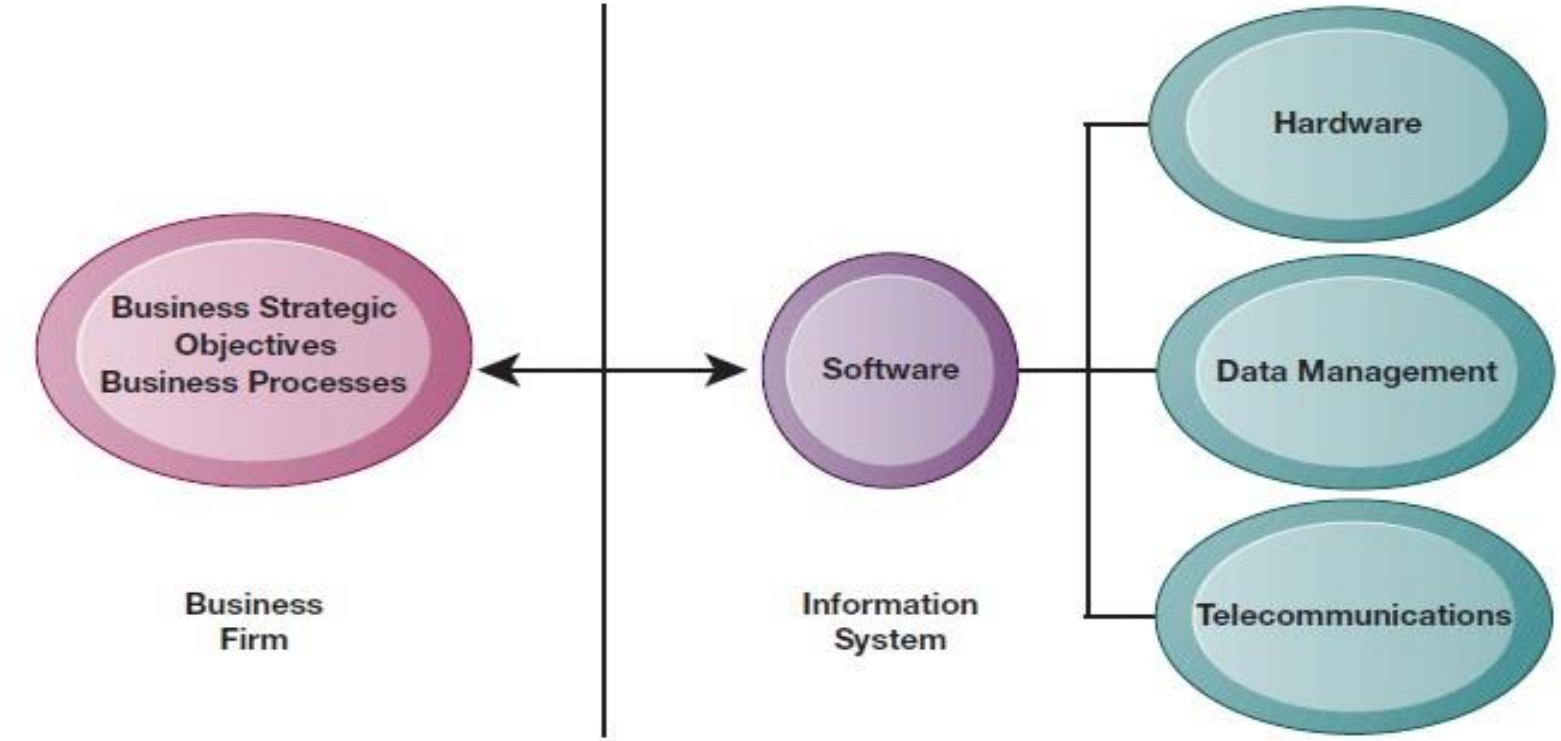
Operations research focuses on mathematical techniques for optimizing selected parameters of organizations, such as transportation, inventory control, and transaction costs.

- **BEHAVIORAL APPROACH**

An important part of the information systems field is concerned with behavioral issues that arise in the development and long-term maintenance of information systems.

Issues such as strategic business integration, design, implementation, utilization, and management cannot be explored usefully with the models used

The Interdependence Between Organizations and Information Systems



Changes in strategy, rules, and business processes increasingly require changes in hardware, software, databases, and telecommunications. Often, what the organization would like to do depends on what its systems will permit it to do.

Summary

- What exactly is an information system? How does it work? What are its management, organization, and technology components?
- How are information systems transforming business and what is their relationship to globalization?
- Why Business Firms Invest Heavily on Information Systems?
- What academic disciplines are used to study information systems? How does each contribute to an understanding of information systems?



Thank you…